



1. A knight begins on the point $(0, 0)$ in the coordinate plane. From any point (x, y) the knight moves to either $(x + 2, y + 1)$ or $(x + 1, y + 2)$. Find the number of ways the knight can reach the point $(15, 15)$.

2. There are sufficiently many items with a price of $\$n.99$ for each nonnegative integer n . A sales tax of 7.5% is applied on all items. If the total cost of a purchase, after tax, is an integer number of cents, find the minimum possible number of items in the purchase.

3. Find the sum of all possible values of n for which it is impossible to divide n students into groups of 12 to 15 members each.

4. There exists a unique positive real number x satisfying

$$x = \sqrt{x^2 + \frac{1}{x^2}} - \sqrt{x^2 - \frac{1}{x^2}}.$$

Given that x can be written as $x = 2^{\frac{m}{n}} \cdot 3^{-\frac{p}{q}}$ for positive integers m, n, p, q , with $\gcd(m, n) = \gcd(p, q) = 1$, find $m + n + p + q$.

5. Let $ABCD$ be a rectangle with sides $AB > BC$ and let E be the reflection of A over $\overline{BD}$. If $EC = AD$ and the area of $ECBD$ is 144, find the area of $ABCD$.

6. Find the number of complex numbers z satisfying $|z| = 1$ and $z^{850} + z^{350} + 1 = 0$.

7. For every positive integer n , define

$$f(n) = \frac{n}{1 \cdot 3 \cdot 5 \cdots (2n + 1)}.$$

Suppose that the sum $f(1) + f(2) + \cdots + f(2020)$ can be expressed as $\frac{p}{q}$ for relatively prime integers p and q . Find the remainder when p is divided by 1000.

8. On day $N = 0$, there is only one person on a planet. For each positive integer $N > 0$, if there is a positive number of people on the planet, then either one of the following three occurs, each with probability $\frac{1}{3}$:

- 1) The population stays the same
- 2) The population increases by 2^N
- 3) The population decreases by 2^{N-1} . (If there are no greater than 2^{N-1} people on the planet, the population drops to zero, and the process terminates.)

The probability that at some point there are exactly $2^{20} + 2^{19} + 2^{10} + 2^9 + 1$ people on the planet can be written as $\frac{p}{3^q}$, where p and q are positive integers such that p is not divisible by 3. Find the remainder when $p + q$ is divided by 1000.

9. Let $ABCD$ be a cyclic quadrilateral with $AB = 6$, $AC = 8$, $BD = 5$, $CD = 2$. Let P be the point on $\overline{AD}$ such that $\angle APB = \angle CPD$. Then $\frac{BP}{CP}$ can be expressed in the form $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

10. Let $1 = d_1 < d_2 < \dots < d_k = n$ be the positive divisors of a positive integer n . Let S be the sum of all positive integers n satisfying

$$n = d_1^1 + d_2^2 + d_3^3 + d_4^4.$$

Find the remainder when S is divided by 1000.

11. An *excircle* of a triangle is a circle tangent to one of the sides of the triangle and the extensions of the other two sides. Let ABC be a triangle with $\angle ACB = 90^\circ$ and let r_A , r_B , r_C denote the radii of the excircles opposite to A , B , C , respectively. If $r_A = 9$ and $r_B = 11$, then r_C can be expressed in the form $m + \sqrt{n}$, where m and n are positive integers and n is not divisible by the square of any prime. Find $m + n$.

12. Define a sequence $a_0, a_1, a_2, \dots$ by

$$a_i = \underbrace{1 \dots 1}_{2^i \text{ 1's}} \underbrace{0 \dots 0}_{(2^i - 1) \text{ 0's}} 1_2$$

where a_i is expressed in binary. Let S be the sum of the digits when $a_0 a_1 a_2 \dots a_{10}$ is expressed in binary. Find the remainder when S is divided by 1000.

13. Aiden writes on a piece of paper the positive integers from 1 to 8 in that order. Then, he randomly writes either $+$ or $\times$ between every two adjacent numbers, each with equal probability. The expected value of the expression he writes can be expressed as $\frac{p}{q}$ for relatively prime positive integers p and q . Find the remainder when $p + q$ is divided by 1000.

14. Let ABC be a triangle with sides $AB = 5$, $BC = 7$, $CA = 8$. Denote by O and I the circumcenter and incenter of $\triangle ABC$, respectively. The incircle of $\triangle ABC$ touches $\overline{BC}$ at D , and line OD intersects the circumcircle of $\triangle AID$ again at K . Then the length of DK can be expressed in the form $\frac{m\sqrt{n}}{p}$, where m , n , p are positive integers, m and p are relatively prime, and n is not divisible by the square of any prime. Find $m + n + p$.

15. Find the number of integer sequences $a_1, a_2, \dots, a_6$ such that

- 1) $0 \leq a_1 < 6$ and $12 \leq a_6 < 18$
- 2) $1 \leq a_{k+1} - a_k < 6$ for all $1 \leq k < 6$
- 3) there do not exist $1 \leq i < j \leq 6$ such that $a_j - a_i$ is divisible by 6.

Time limit: 180 minutes